William Wu

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Education

University of Waterloo Sep 2024 – Present
Bachelor of Computer Science, Honours *Waterloo, ON*

- Cumulative average of **87.4%** and one of ten **Mathematics National Scholarship** recipients

Technical Skills

Languages: Go, C++, Python, C, Java, JavaScript, TypeScript, SQL, Bash, R, HTML/CSS
Frameworks: React, Next.js, Gin, FastAPI, Express, Flask, Node.js, Tailwind CSS
Technologies: AWS (*AWS Certified Cloud Practitioner*), Docker, Linux, PostgreSQL, gRPC, Temporal, Git

Experience

Roblox | *Software Engineer Intern* May 2026 – Aug 2026
Go, React, gRPC, Temporal *San Mateo, California*

- Owned **end-to-end development** of **trademark infringement reporting and resolution** in Roblox Creator Hub
- Architected **extensible** trademark structures across existing IP **DB schemas, Go models, and API handlers**
- Extended existing **Temporal workflows** within **IP microservices** to enable durable trademark claim processing
- Built **creator-facing UI and internal IP moderation tools**, aligning requirements across **legal, ops, and design**

LeapAP | *Software Developer Intern* May 2025 – Aug 2025
JavaScript, Node.js, Express *Aurora, Ontario*

- Engineered robust **web scraping** solutions with **Puppeteer**, improving document collection reliability and coverage
- Constructed **regex**-based parsers with **95%+ accuracy** in extracting critical data from unstructured documents
- Leveraged **Grafana and Graylog** to monitor and resolve scraping anomalies, ensuring **under 2.5% error** rates
- Accelerated development for **15+ scrapers** by introducing reusability and refactoring the scraping framework

WATonomous | *Autonomous Software Developer* Feb 2025 – Aug 2026
C++, Python, ROS 2, Docker *Waterloo, Ontario*

- Implemented a **planner node** for a robotics simulation, utilizing the **A* algorithm** to process costmap data
- Integrated a **YOLOv8 model** into a Mars rover and **fine-tuned** it for object detection using RealSense cameras
- Designed and incorporated realistic terrain and props in a Mars rover **Gazebo** simulation using **mesh assets**

DMOJ Modern Online Judge | *Contest Organizer* May 2021 – Feb 2023
C++, Java, Python, LaTeX *Remote*

- Applied **advanced algorithms and data structures** to **create 9 contests**, with a total of **1,500+ live participants**
- Authored **12 competitive programming problems** with a combined total of **13,000+ user submission attempts**
- Engineered black-box tests for **15+ rated programming contests**, preventing unintended edge-case solutions

Projects

Custom Terminal Text Editor | *C++, OOP, STL, ncurses*

- Engineered a modal text editor using the **MVC architecture** with syntax highlighting and custom keybindings
- Leveraged **RAII and smart pointers** to ensure memory safety and eliminate manual memory management

Re-Prompt-That: Multiplayer AI Prompting Game | *JavaScript, Next.js, Node.js, Socket.IO*

- Crafted a cross-platform multiplayer experience by integrating a **WebSocket server** with a seamless frontend
- Optimized event logic, reducing game start **latency to under 50 ms** and enabling tight synchronization for users

Layoff Evaders: VR Fitness Game Companion App | *Python, Firebase, Pandas, Streamlit, Cohere*

- Achieved **2nd place overall** out of **600+ competitors and 130+ submitted projects** during DeltaHacks XI
- Developed a real-time session statistics dashboard using **Firebase and Cohere's LLM** to enhance interactivity

AdmissEd: Admission Statistics Web App | *AWS, React, Python, FastAPI, Pandas, Matplotlib*

- Deployed *admissed.com* by employing AWS cloud technologies such as **S3, Lambda**, and the **CloudFront CDN**
- Built a responsive full-stack web app utilizing React, featuring a **RESTful FastAPI backend** for data parsing
- Harnessed the **Pandas and Matplotlib** data analysis and visualization libraries to generate in-depth reports

Achievements

USA Computing Olympiad | *Platinum Division* - Promoted by scoring **746/1,000** on the Gold Contest Dec 2022

Canadian Computing Competition, Senior | *Honour Roll* - Placed **75th out of 3,262 nationally** Feb 2022